TOGETHER

Together Empowering App and IoT-Based Smart Farming System

Executive Summary

Climate change has a profound impact on agriculture, as it depends on climate and environmental conditions. It poses a major threat to crop yields by increasing the spread of pests and diseases, resulting in annual losses of up to 40% of crop production worldwide. This, in turn, costs the global economy billions of dollars and poses a significant threat to global food security. Weeds and crop diseases are the most common affecting issues that cause crop loss. Today losses caused by weeds and insects can reach 40% of total crop yields each year and this percentage is expected to increase significantly in the coming years.

In our daily routines, we often overlook the impact our habits can have on the environment. Every choice we make can contribute to climate change; from the energy we consume to the waste we produce. Simple actions like leaving lights on when unnecessary, using single-use plastics, and wasting water contribute to increased carbon emissions and environmental degradation. The lack of awareness about these impacts further aggravates our challenges in combatting climate change.

We are proposing **Together,** a comprehensive solution that addresses two main challenges posed by climate change, the crop yields losses in the agriculture sector and the lack of awareness about climate change among the communities. It comprises an empowering mobile app that provides a tool for users to help them normalize planting around with a platform to share information and experiences, and a smart farming system that will help mitigate the effects of climate change on the agriculture sector and increase the productivity.

The empowering mobile app introduces several innovative features such as :

- Smart Planting Guide, powered by deep learning models to predict the plant types based on leaves and flowers to equip the users with how to care instructions for successful planting process.
- Community Platform that fosters climate change awareness by facilitating information sharing.
- Eco-friendly products Marketplace.
- The Pollution Reporting feature empowers users to report polluting activities.

The smart agriculture system provides three AI-driven services which are :

- The Weed Detection system efficiently identifies weeds at different growth stages.
- The Disease Detection system leverages the AI power to identify diseases in rice crops.
- The IoT-Based Crop Recommendation and Support system predicts suitable crops based on soil analysis and with help of the Nutrients Helper, the user gets the required amounts of soil enhancers, utilizing the crop management practices.

Our goal is to build more aware communities and provide more efficient and green agriculture solutions to reach net-zero emissions and achieve the maximum productivity.

Here are the key features that we dedicated our efforts to bring to light in our project :

1- Smart Planting Guide:

The deep learning-based Smart Planting Guide is a groundbreaking feature that revolutionizes how users approach plant care. By accurately predicting and recognizing plant types through leaves and flowers, and with the help of our plants database, users receive tailored care instructions for each plant in their garden. This empowers even novice gardeners to nurture plants effectively.

2- Community Platform:

The Community Platform is a pioneering concept that leverages social connectivity to raise climate change awareness. By providing a space for information sharing, experiences, and sustainable practices, the app fosters a sense of environmental responsibility. The inclusion of an eco-friendly product marketplace incentivizes eco-conscious consumer behavior that guides the users towards local brands producing eco-friendly products and gardening supplements. Along with a reward program to collect points for their contributions on the platform to redeem those points with products. We consider this an innovative approach for promoting sustainable behavior through our app.

3- Pollution Reporting:

The Pollution Reporting feature is a unique addition to the app, allowing users to report polluting activities effortlessly. By capturing images and providing location information, users become environmental watchdogs. This innovation facilitates prompt action by specialized authorities, creating an avenue for community-driven environmental protection. The merit lies in empowering individuals to actively contribute to safeguarding their environment.

4- Intelligent Weed Detection:

The smart agriculture system's Intelligent Weed Detection feature is a significant breakthrough in weed management. Through fine-tuned pre-trained models, it identifies and classifies weeds at different growth stages, and using the collected database, the system offers a suitable treatment. Early detection allows farmers to implement timely interventions, minimizing crop losses and increasing yields. The merit lies in improving crop monitoring and increasing productivity to ensure global food security.

5- Intelligent Disease Detection:

The Intelligent Disease Detection system is a vital component of the smart agriculture system, accurately identifying diseases in rice crops during early stages. By employing transfer learning and a fine-tuned ResNet34 model, farmers can swiftly contain diseases and prevent widespread crop losses. This innovation's merit lies in bolstering sustainable agricultural practices and supporting food security.

6- IoT-Based Crop Recommendation and Support:

The IoT-based Crop Recommendation and Support system revolutionizes crop management through data-driven insights. Predicting suitable crops based on climate variables and soil nutrients optimizes farmers' decision-making processes. Additionally, the system offers nutrient recommendations, promoting sustainable practices and minimizing environmental impact. The merit lies in providing farmers with precision agriculture techniques, leading to increased productivity and resource utilization.

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1. Introduction and Background Review

1.1 Introduction

In a world grappling with the far-reaching impacts of climate change, the agriculture sector stands particularly vulnerable. The profound consequences of climate change, including rising temperatures, irregular weather patterns, and increased pests and diseases, pose formidable challenges to farmers worldwide. As a result, reduced crop yields threaten the availability and affordability of food for communities around the globe. The need to address these pressing issues and safeguard global food security has never been more urgent as food security remains a pressing global concern, and agriculture plays a pivotal role in addressing it. With a rapidly growing population, ensuring a stable and sustainable food supply is paramount. However, climate change-induced challenges in agriculture threaten to undermine global food security, emphasizing the need for innovative solutions.

Another obstacle to combating climate change is the lack of awareness among individuals. Many people need to be made aware of the profound impact their daily choices can have on the environment. Raising awareness and encouraging sustainable practices are pivotal in fostering collective climate action. In light of these challenges, the "Together" project aims to provide transformative solutions. By integrating cutting-edge technologies into both the mobile app and the smart agriculture system, we seek to empower farmers and individuals to make informed decisions, optimize crop management, and contribute to climate change mitigation efforts.

Through the "Together" project, we envision a future where agriculture is more resilient, sustainable, and attuned to the changing climate. Our innovative ideas and features, backed by AI and IoT technologies, offer promising avenues for fostering sustainable practices, enhancing crop yields, and raising climate change awareness. We are committed to driving positive change and making a meaningful impact on the agriculture sector and our planet as a whole.

1.2 Background Review:

We conducted a literature review on climate change, agriculture, deep learning, and related tools and techniques to raise awareness. We reviewed several research papers, articles, and online resources to gain a better understanding of the subject matter. We also studied existing applications and systems that address climate change and plant caring. Researchers have attempted several methods to classify and diagnose plant diseases and extract their features. Deep learning alongside image processing and traditional machine learning techniques have been used in the agricultural field. This section concentrates on some of the previous work discussing the rice crops diseases, weed classification, leaves classification, flowers classification and crop recommendation systems using soil analysis.

In [1], a semi-automatic Object Based Image Analysis (OBIA) procedure has been developed with Random Forests (RF) combined with feature selection techniques to classify soil, weeds, and maize. With all these articles, we can notice that the selected features change in general from one type of culture to another or from one type of data to another.

And in a recent study, Di Cicco et al. [2] suggested the use of synthetic training datasets. However, this technique requires precise modeling in terms of texture, 3D models and light conditions. Currently, most methods used for unsupervised data collection detect crop rows first, then label plants within the

rows as crops and those between rows as weeds (inter-row weed) [3]. These methods strongly depend on the presence of weeds in the inter-row. Therefore, labeled data will be very unbalanced in cases where the field has fewer weeds between crop rows. This can reduce the efficiency of the trained models and promote overfitting.

However, such traditional image augmentation techniques and transfer learning provide highly correlated images and only little additional information to the deep-learning model. This not only reduces the ability of the model to generalize but leads to over-fitting problems. Several researchers have proposed an automated identification system for rice disease detection. In [4], rice disease detection was accomplished by considering the area of the diseased leaf using image processing and a model was developed using the Naive Bayes Classifier. This classifier could identify three types of diseases of paddy plants with a prediction accuracy of 89%.

Lei Feng et al. [5] employed hyperspectral imaging(HSI) to detect paddy leaf diseases and developed a CNN architecture as a classification model using deep transfer learning techniques. They found fine-tuning was the most efficient solution, achieving 88% accuracy.

The field of smart farming heavily relies on the integration of machine learning and IoT technologies, but implementing these technologies comes with challenges. Soil fertility is critical to maintaining crop production levels, but the nutrient levels of soil decrease over time due to cultivation. One method to improve soil fertility is by optimally increasing it using sensor technology, which can improve soil quality, food safety, and crop profitability.

The excessive use of chemical fertilizers imbalances the availability of soil nutrients, which are collected, classified, and can be analyzed using an extreme machine learning decision system. Hence, we should avoid a deficiency in NPK fertilizer in plants, as it might lead to bad results. Excess usage of fertilizer imbalances ecosystems.

Shubham Prabhu, et.al. [6] has proposed a paper in which they put forward the soil analysis and crop prediction model. The main aim of the paper is to create a prediction engine that will be for the most suitable crop for a particular soil. Basically, the author considered five soil samples from different regions and then analysis is done based on temperature, moisture and humidity which is carried out at a regular interval of 24 hours and the data is uploaded, displayed. All the data that is analyzed is continuously monitored, displayed, and uploaded on the IoT cloud. They have basically used three algorithms Naïve Bayes, Logistics and C4.5 [7] in which has given accuracy of 85% .

As machine learning technology advances, increasingly sophisticated models have been proposed for automatic plant identification. With the widespread use of smartphones and the emergence of mobile apps like PlantNet [8], millions of plant photos have been collected. Mobile-based automatic plant identification is essential for real-world applications such as social-based ecological surveillance [9], invasive exotic plant monitoring [10], ecological science popularization, and more.

In recent years, many efforts have been made to extract local characteristics of leaves, flowers, or fruits. Most researchers use variations in leaf characteristics as a comparative tool for studying plants, and some leaf datasets, including the Swedish leaf dataset, Flavia dataset, and ICL dataset, are considered standard benchmarks. In [11], Soderkvist extracted shape characteristics and moment features of leaves and analyzed 15 different Swedish tree classes using backpropagation for the feed-forward neural

network. In [12], Zhanget al. combined Harr features with SIFT features of flower image, coding them with non-negative sparse coding method and classifying them by k-nearest neighbor method.

After so many years of continued exploration into plant recognition technology, the dedicated mobile applications such as LeafSnap [13], Pl@ntNet [8], can be conveniently used for identifying plants. In [14] the author has used a deep residual network that uses a residual map to identify the pest, including improved computational frameworks, specifically Graphical Processing Units (GPU) embedded processors. The author used DL to identify plant diseases. A deep convolutional neural network was used in this research. CNN, RNN, and GAN are the most common Deep Learning algorithms.

Finally, after this literature survey, we found that creating awareness about climate change is a complex issue, and traditional methods have not delivered fast enough results. Mobile applications have shown promise in promoting awareness and motivating individuals to take small actions that contribute to reducing climate change. AI and IoT technologies can be used to provide weed and disease detection systems, crop recommendations, and plant type identification based on leaves and flowers features.

2. System Description

2.1 Needs Assessment

This needs assessment aims to identify the critical requirements and challenges that the system intends to address. The empowering app is designed to provide users with a comprehensive tool to make a positive impact on the environment by making planting easier, promote sustainable practices, and contribute to the fight against climate change. The following assessment outlines the key needs that the app aims to fulfill:

- **Raising Awareness about Climate Change**
  There is a significant need to raise awareness about climate change. Many individuals lack access to reliable information and resources related to climate change, leading to misconceptions and a sense of hopelessness regarding individual contributions. The empowering app aims to fill this gap by offering a platform that educates users about the challenges posed by climate change and the potential solutions they can undertake.
- **Empowering Individuals to Act**
  Individuals often underestimate the impact of their actions in combating climate change. The app seeks to empower users by providing tangible ways to contribute positively. Through its AI-powered smart planting guide, users can normalize planting around them and make planting easier, promoting green spaces and carbon sequestration.
- **Facilitating Community Engagement and Sharing Experiences**
  The app recognizes the power of community engagement and the impact of collective efforts. It addresses the need for a platform where individuals can come together, share their experiences, and inspire others to adopt sustainable practices. The community platform in the app encourages users to collaborate, exchange ideas, and support each other in their journey towards environmental conservation.
- **Reporting and Addressing Pollution Activities**
  One of the pressing needs is to address pollution activities, such as the burning of rice straws, which significantly contribute to air pollution and climate change. The app's pollution reporting

feature allows users to report such activities, helping authorities and environmental organizations take appropriate action to mitigate the impact.

- **Motivating Behavioral Change**
  The app seeks to instill a sense of responsibility and motivation among users to make conscious decisions that positively impact the environment. By providing real-time feedback, rewards, and recognition for sustainable actions, the app encourages behavioral change and fosters a culture of environmental stewardship.

For the smart farming system, it should be designed to provide farmers and agriculture sector with a tool to mitigate the effects of climate change on the agriculture sector and decrease crop losses.
The following assessment outlines the key needs that the system aims to fulfill:

- **Addressing Weed Infestation**
  One of the primary needs of the smart farming system is to address the issue of weed infestation in crop fields. Weeds compete with the main crop for essential resources and can significantly impact crop growth and yield. The intelligent weed detection system addresses this need by identifying and classifying different weed species at various growth stages. This capability allows farmers to take timely and targeted action to control weeds, thereby minimizing their negative impact on crop productivity.
- **Early Detection and Management of Crop Diseases**
  The smart farming system fulfils the crucial need for early detection and management of crop diseases in rice crops. Identifying diseases at their initial stages is essential to prevent their spread and to avoid significant losses in crop yield. This early disease detection empowers farmers to apply appropriate treatments and containment measures promptly, ensuring healthier crops and improved overall yield.
- **Utilizing IoT and Data-Driven Insights**
  The integration of IoT sensors and data-driven insights is a fundamental aspect of the smart farming system. The use of IoT sensors to monitor soil nutrients, climate variables, and other environmental factors empowers farmers with precise data, enabling them to optimize crop selection, irrigation, and fertilization practices.
- **Customized Crop Recommendations**
  One of the critical needs of modern agriculture is the ability to provide farmers with customized crop recommendations based on specific soil and climate conditions. The IoT-based crop recommendation and support system predicts the most suitable crops for each season, considering factors such as temperature, soil pH, NPK values and humidity. This tailored approach ensures that farmers can optimize their crop choices, leading to increased productivity and reduced risk of crop failure.
- **Cost-Effective and Sustainable Farming Practices**
  The smart farming system aims to promote cost-effective and sustainable farming practices. By utilizing deep learning models and transfer learning, the system minimizes the need for manual labour and chemical interventions, making it a more environmentally friendly and economically viable solution for farmers. The data-driven recommendations for soil health and nutrient management also enable farmers to use fertilizers and soil enhancers efficiently, reducing wastage and potential environmental harm.

2.2 Functional requirements

Empowering app

- **Login and sign up**
  Users should be able to create an account and log in to access the app's features.
  Users should be able to reset their password if they forget it.
- **User Profile**
  Users should be able to create and edit their profile, including personal info.
  Users should be able to view other users' profiles and follow them.
- **Community Features**
  Users should be able to post updates, comment, like, share photos and videos.
- **Plant Identification and caring guide**
  Users should be able to take photos of leaves and flowers and receive identification results.
  The identification system should use image recognition algorithms to identify plant species.
  Users should be able to see information about the plant and add it to their garden list.
- **Eco-friendly Marketplace**
  Users should be able to browse, make an order, and purchase eco-friendly products, such as biodegradable packaging, organic fertilizers, and gardening supplies.
- **Pollution Reporting**
  Users should be able to take pictures and send a pollution report with the location coordinates.

Smart farming system

- **User Registration and Authentication**
  Farmers and authorized users should be able to register and create accounts on the platform.
- **Intelligent Weed Detection**
  Farmers should be able to upload images of crop fields to the system for weed detection.
  The system must process the images and use the models to classify weeds accurately.
  Detected weed species and their treatment should be displayed on the user dashboard.
- **Intelligent Disease Detection**
  Farmers should be able to upload images of rice crops to the system for disease detection.
  The system should utilize the model to identify and diagnose diseases in the early stages.
  Detected diseases and corresponding treatment recommendations should be provided on the user dashboard.
- **IoT Sensor Data Collection**
  The system must integrate with IoT sensors placed in the soil to collect real-time data on temperature, soil pH, NPK values, humidity, and rainfall rate.
  Collected sensor data should be transmitted securely to the backend cloud server for processing.
- **Crop Recommendation and Support System**
  Based on the collected sensor data, the system should recommend the most suitable crops for each season.
  Data-driven recommendations for optimal nutrient and soil enhancers should be provided to improve crop yield.

- **Dashboard and Data Visualization:**
  The system should offer an interactive and user-friendly dashboard for farmers to view real-time data insights, analysis, and recommendations.
  Visual representations of weed and disease detection results, sensor data, and crop recommendations should be presented on the dashboard.

2.3 Non-functional requirements

- **Usability**: The application should be user-friendly, with clear navigation and intuitive interfaces. It should be easy for users to access the features and functions they need, with minimal training.
- **Performance**: The app should perform quickly and efficiently, with minimal loading times and delays. It should be able to handle large amounts of data and traffic without crashing.
- **Reliability**: The application should be reliable and stable, with minimal downtime or errors. It should be able to recover from failures quickly and without data loss.
- **Security**: The app should be secure, with robust authentication and encryption mechanisms to protect user data and prevent unauthorized access.
- **Scalability**: The application should be scalable, able to handle an increasing number of users and data without significant performance degradation or downtime.
- **Compatibility**: The app should be compatible with a wide range of devices and operating systems, including smartphones, tablets, and desktop computers. It should also work seamlessly with other applications and services.
- **Accessibility**: The application should be accessible to users with disabilities, with features such as screen reader compatibility, adjustable font sizes, and high contrast modes.

2.4 Economical Constraints:

- **Budget Limitations**: We had to carefully manage expenses and allocate resources within the predetermined budget as the inflation wave has crested while developing our project. This includes costs related to software development, hardware infrastructure, data storage, cloud services, and any necessary third-party services.
- **Cost-effective Solutions**: To ensure efficiency and accuracy, the team prioritized cost-effective technologies and tools. We have used open-source software and readily available APIs and platform to avoid expensive proprietary solutions.
- **Affordability for Farmers**: The system should be cost-effective for farmers, particularly small-scale and resource-constrained farmers, to encourage widespread adoption.

2.5 Technical Constraints:

- **Mobile App Compatibility**: The mobile app must be designed to work seamlessly on various smartphone operating systems, including iOS and Android. Extensive testing and optimization are necessary to ensure a smooth user experience across different devices and screen sizes.
- **Hardware and Connectivity:** The technical implementation must consider the availability of suitable hardware and reliable internet connectivity in rural farming areas. Limited internet access or unstable connections may impact data transmission and real-time functionalities.
- **Data Security**: Both the mobile app and webpage should adhere to robust data security protocols to protect user information and prevent unauthorized access. Encryption methods and secure authentication mechanisms must be implemented to safeguard sensitive data.

2.6 Logistical Constraints:

- **Data Collection**: Acquiring large-scale and diverse datasets for training deep learning models might be challenging due to the diversity of crops, pests, and diseases across different regions.
- **Drone Acquiring:** We faced a lot of problems and paperwork to acquire the drone. These regulations involved obtaining permits, licenses, and adhering to specific flight restrictions. We are still on it to complete our vision for our project.
- **Sensor Deployment**: The deployment and maintenance of IoT sensors in the field might face logistical challenges, including sensor calibration, battery replacement, and weatherproofing.
- **Scalability:** The system's architecture should be designed to accommodate future growth and increasing user demands. Scalability constraints may arise when scaling up the system to serve a larger user base.

3. Overall System Architecture

The system comprises two solutions: an empowering mobile app and a web-based smart farming system. The app allows users to engage through a social platform, access a planting guide, and report pollution incidents. User-generated imagery data is sent to the cloud, where AI models perform plant recognition, weed detection, disease detection, and crop recommendations.

Here is our high-level architecture showing the dataflow for the app and the farming system:

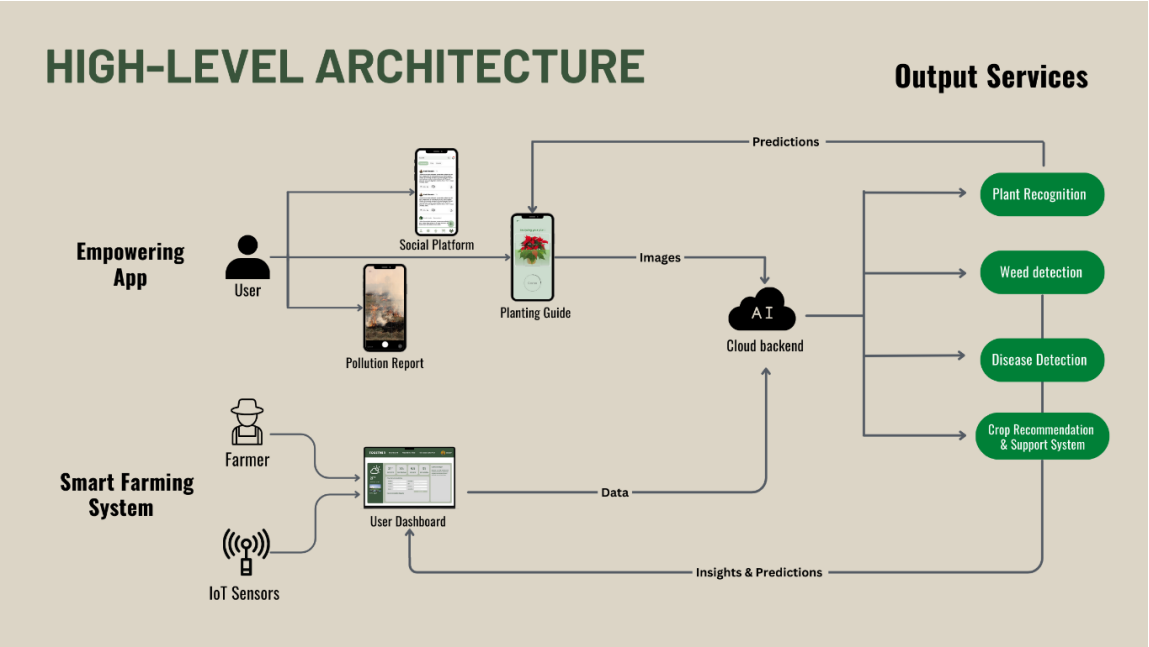


Fig. 1 System High-level Architecture Diagram

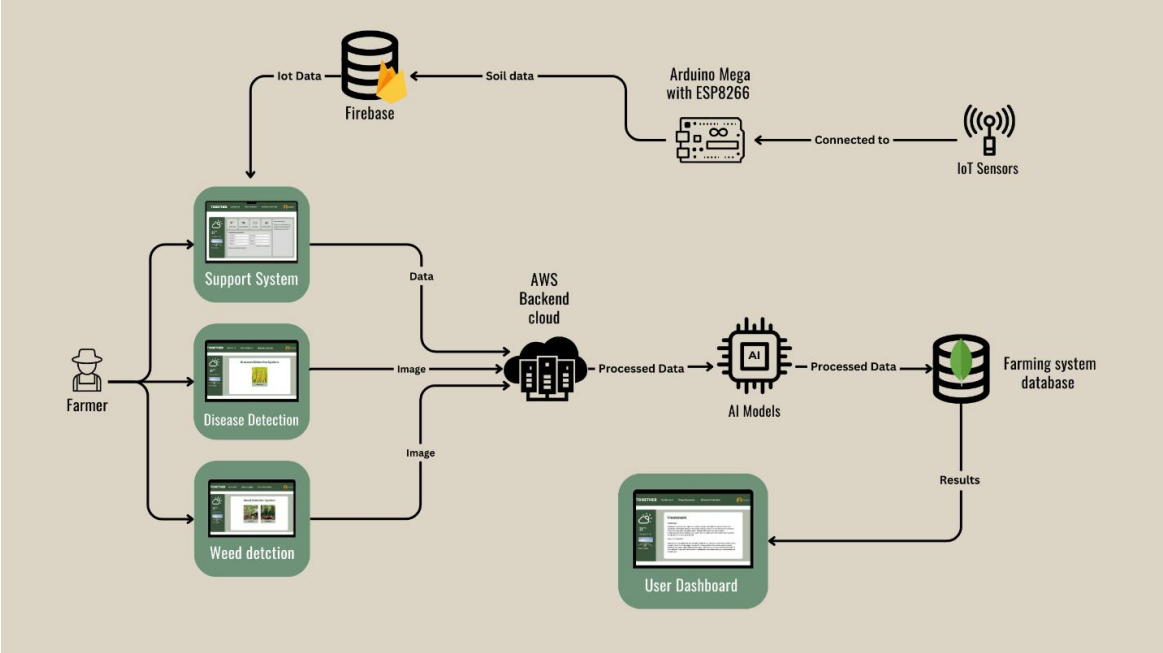


Fig. 2 Empowering App Architecture Diagram

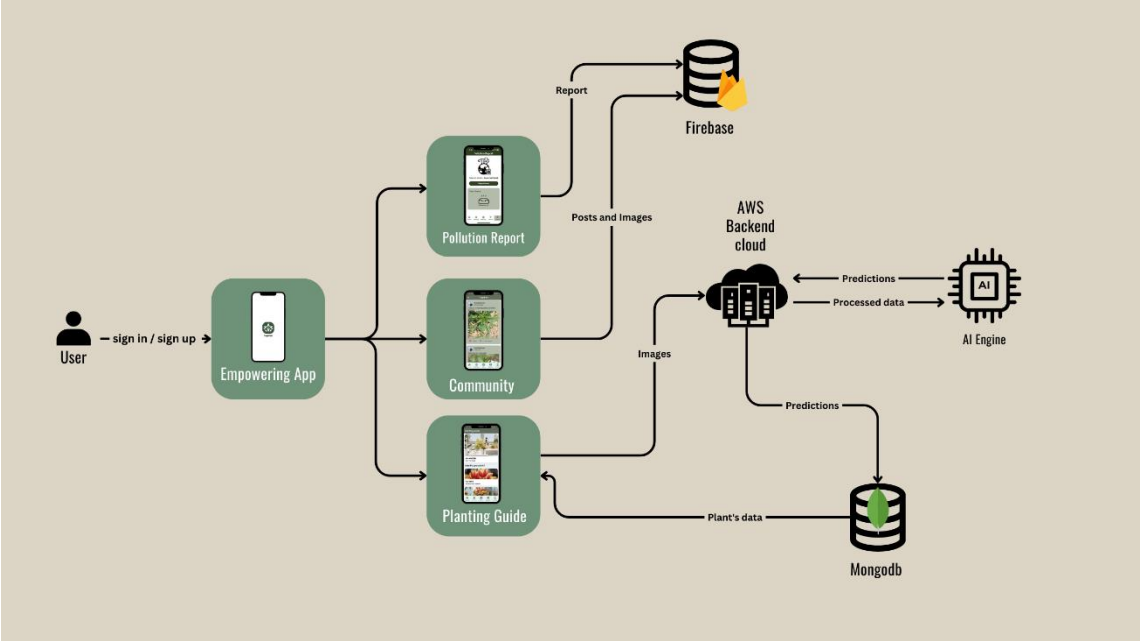


Fig. 3 Smart Farming System Architecture Diagram

Hardware

For the hardware, we have used Arduino mega board, connected to 7 sensors which are, DHT11 temperature & humidity sensor, soil pH , soil moisture , NPK sensor along with the Esp8266 module to send the data over network. We have used Max485 bus to connect the NPK and soil pH sensors to the Arduino board. The data is sent to the firebase real-time database to be stored.

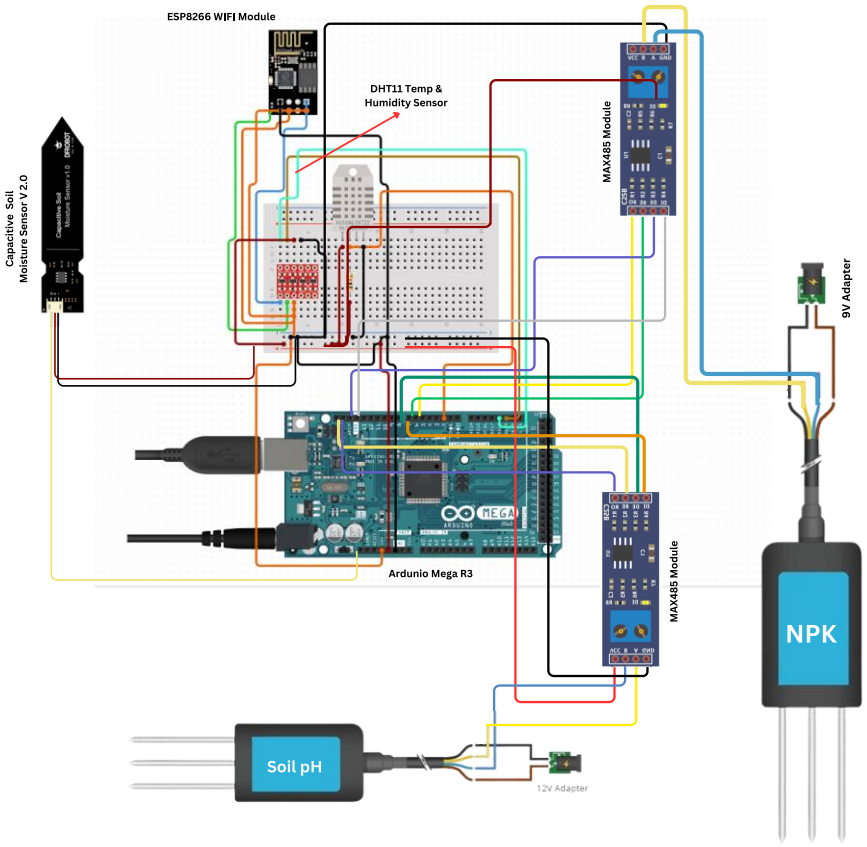


Fig. 4 IoT system diagram

4. Implementation Details

This section describes some main concepts and methods used in the proposed models and applications. We have implemented the whole project through **3** main phases.

4.1 Data Acquisition

A collection of **six** datasets that are involved which are: flower classification, leaf classification, cotton weeds, crop recommendation data, paddy diseases and weed seedlings, to satisfy our model's training needs.

1- Flowers Dataset [15]

The flower classification dataset is a collection of images of flowers from 16 different species. The dataset contains a total of 15740 images, with between 800 to 1027 images per species. Each image is labeled with the species it belongs to, making it a supervised learning problem for classification.

2- Leaves Dataset [16]

The Leaf Classification dataset consists approximately 1,584 images of leaf specimens (16 samples each of 99 species) which have been converted to binary black leaves against white backgrounds. Three sets of features are also provided per image: a shape contiguous descriptor, an interior texture histogram, and a fine-scale margin histogram. For each feature, a 64-attribute vector is given per leaf sample.

3- Paddy Diseases Dataset [17]

The Paddy Doctor Dataset provides a training dataset of 10,407 (75%) labeled paddy leaf images across ten classes (nine diseases and normal leaf). We also provide additional metadata for each image, such as the paddy variety and age. Our task is to develop an accurate disease classification model using the training dataset and then classify each sample in the test dataset of 3,469 (25%) paddy leaf images into one of the nine diseases or normal leaf.

4- Seedlings Dataset [18]

The plant seedling classification dataset contains a collection of images of plant seedlings from 12 different species at various stages of growth. The dataset is organized into a training set and a test set, with each image assigned a unique filename. The 12 different species divided as weeds and crops, 8 seedlings correspond to wild weeds like: loose silky-bent, common chickweed, scentless mayweed, small-flowered cranesbill, fat hen, charlock, cleavers, black-grass, and shepherd's purse and 4 seedlings that belong to common agriculture crops like sugar beet, maize, and common wheat at various stages of growth. The dataset contains a total of 5,539 images in the training set and 7,303 images in the test set, with each image labeled with the species it belongs to.

5- Crop Recommendation Dataset [19]

The Crop Recommendation dataset is a collection of data related to crop farming. The dataset includes information on different parameters such as soil content, temperature, humidity, and rainfall, which can be used to make informed decisions about farming strategy. The goal is to build a predictive model that can recommend the most suitable crops to grow in a particular farm based on these parameters. The dataset was constructed by aggregating datasets of rainfall, climate, and fertilizer data from India.

6- Cotton Weeds Dataset [20]

The dataset CottonWeedID15 consists of 5187 RGB images of 15 weeds that are common in cotton fields in the southern U.S. states. These images were acquired by either smartphones or hand-held digital cameras, under natural field light condition and at varied stages of weed growth in 2020 and 2021. The images were manually labeled by weed scientists and trained individuals. The images are in JPEG format and have a resolution of 256 x 256 pixels.

4.2 Preprocessing Data & Training the Models

After choosing the suitable dataset for each feature, we moved forward to build and train the deep learning models, starting with the **IoT-Enabled Smart Farming System :**

To develop the **Intelligent weed detection system**, we have started to prepare our data. We had to prepare two datasets, the Cotton Weeds dataset, which consists of 15 weeds and the Plant Seedlings Dataset, consists of plant seedlings from 12 different species divided as 8 seedlings for weeds and 4 seedlings that belong to common agriculture at various stages of growth.

- **Data Preprocessing**

For the Cotton Weeds dataset, we have used data augmentation, which is a technique used in machine learning and computer vision to increase the diversity of a training dataset. This is achieved by applying various transformations to the images or data samples, such as rotation, flipping, cropping, and color changes, in order to create new variations of the original data. These augmented samples can then be used to train machine learning models, improving their ability to generalize to new, unseen data.

We have used a custom augmentation function (see **Fig**. 5), that applies a series of data augmentations to each image in the dataset. The augmentations are defined using the transforms module from PyTorch, and include random rotations, flips, and affine transformations, as well as random cropping with size (224, 224) and color jittering. The RandomApply function is also used to apply certain augmentations with a probability p, allowing for additional randomness in the dataset.

```
transforms.RandomApply([transforms.RandomCrop(size=(224, 224), padding=20)], p=0.3),
transforms.RandomApply([transforms.GaussianBlur(kernel_size=3, sigma=(0.1, 2.0))], p=0.2),
transforms.RandomApply([transforms.RandomPerspective(distortion_scale=0.2)], p=0.2),
```

Fig 5. Code snippet shows the RandomApply function.

For the Plant Seedlings dataset, we have used data augmentation, with, such as horizontal flipping, shearing, scaling, translation, rotation, and brightness shift, to enhance the training process. These augmentations aimed to diversify the training data, improve the model's ability to handle variations in object orientation, shape, position, lighting conditions, and enhance its generalization. The ImageDataGenerator from TensorFlow was employed for data loading and augmentation operations, resulting in improved performance and robustness of the model.

- **Building & Training the Models**

After preparing the data of both datasets, the authors have developed two transfer learning classifiers, a fine-tuned ResNet50 deep learning model for the task of classifying cotton weeds into 15 categories and EfficientNetB2 deep learning model for classifying the plant seedlings.

- **ResNet50**

ResNet50 is a powerful convolutional neural network architecture that has been widely used in computer vision tasks due to its ability to achieve state-of-the-art performance on a variety of datasets. It uses the Residual Connections approach, which avoids "gradient vanishing" and speeds up the training. ResNet50 builds upon ResNet34 by utilizing a deeper network architecture to achieve higher accuracy without encountering the vanishing gradient problem.

The ResNet50 architecture consists of 50 layers that function hierarchically, with the initial layers extracting low-level features and the deeper layers learning more complex, high-level features. Feature extraction is achieved using convolutional layers, batch normalization, and the Rectified Linear Unit activation function. The first layer is a convolutional layer with a 7x7 kernel size and a stride of 2, followed by a max-pooling layer with a 3x3 kernel size and a stride of 2 **(**see **Fig.** 6**)**.

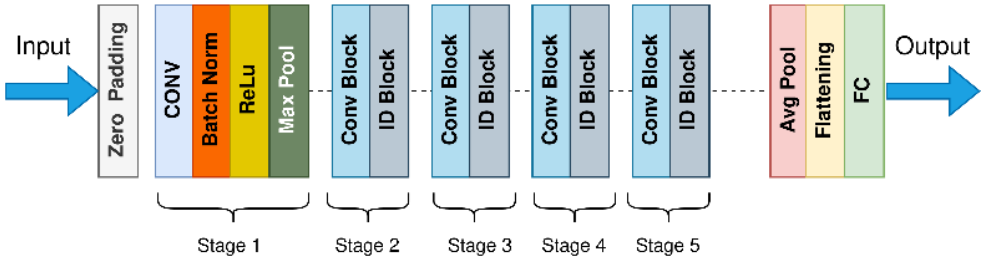


Fig 6. Resnet50 architecture (Wikimedia Commons)

For the training phase, a PyTorch Dataset object has been used, which can be passed to a Data Loader for efficient loading and batching of the data. During the classification stage, the ResNet50 network uses the extracted features to classify the input image into one of 15 categories, to identify the weeds. The features are passed through fully connected layers to produce a probability distribution over the feasible categories, with the category having the highest probability being designated as the predicted class of the image. Our implementation of the ResNet50 model was built using the PyTorch Lightning framework, which provides a high-level interface for training and evaluating deep learning models. The model was initialized with weights pre-trained on the large-scale ImageNet dataset, which contains over 1 million labeled images across 1,000 categories. This pre-training allowed the model to learn a set of general features that could be fine-tuned for the specific task of weed classification.

The ResNet50 model consists of multiple layers of convolutional and pooling operations, followed by a fully connected layer that produces the final predictions. We replaced the original fully connected layer with a new one consisting of two linear layers and a dropout layer in between, with the final output layer having the same number of neurons as the number of classes in our dataset (i.e., fifteen). We used the cross-entropy loss function to calculate the difference between the predicted and actual labels during training. We used the Adam optimizer with a learning rate of 3e-4 and a StepLR learning rate scheduler, which reduces the learning rate by a factor of 0.1 every two epochs. We used a stratified k-fold cross-validation approach with 4 folds to evaluate the model's performance on our dataset.

K-fold cross-validation is a technique that helps in assessing the performance of a model by dividing the dataset into K subsets or folds of approximately equal size. The K-fold cross-validation method involves splitting the data into K parts, where K-1 parts are used for training the model, and the remaining one part is used for testing the model's performance. This process is repeated K times, with each subset used as the testing data once. The results are then averaged across the K experiments to get a final estimate of the model's performance.

The K-fold cross-validation is that it helps in providing a more accurate estimate of a model's performance. By using multiple splits of the data, we can get a better sense of how well our model generalizes to new data. K-fold cross-validation is used to evaluate the ResNet50Model. The dataset is split into 4 folds using the Stratified Fold function from scikit-learn. For each fold, a new instance of the ResNet50Model is created, and the model is trained on the training subset and evaluated on the validation subset using the PyTorch Lightning Trainer.

During training, we monitored the loss and accuracy metrics on both the training and validation datasets and used early stopping with a patience of two epochs to prevent overfitting. We also used the PyTorch Lightning metric module to calculate the accuracy of the test dataset after training was complete.

- **EfficientNetB2**

It is a convolutional neural network architecture with a scaling method that uniformly scales depth, width, and resolution using a compound coefficient. This scaling approach enables efficient utilization of computational resources by uniformly adjusting network depth, width, and image size.

The images were organized into a directory structure where each species had its own subdirectory containing the images. The dataset was split into a training set (80% of the images) and a validation set (20% of the images). We used Keras' ImageDataGenerator to apply data augmentation techniques such as rotation, shifting, shearing, zooming, and flipping to the training set images. This augmented data was used to train the model and improve its generalization performance. The model building process focused on implementing an efficient model to classify the weeds and crops seedlings with high accuracy.

The model's top layers were excluded to allow for customization. A global average pooling layer was added to obtain a fixed-length representation of the feature maps. To construct the predictor layers, a series of fully connected (Dense) layers were added, with ReLU activation functions for non-linearity. Dropout regularization was incorporated to prevent overfitting. The final layer consisted of a Dense layer with a softmax activation function, producing probabilities for each of the 12 target classes.

The model was compiled using the Adam optimizer, Categorical Crossentropy loss function, and Categorical Accuracy as the evaluation metric. The model was trained using the provided training and validation data generators with a batch size of 32 for a total of 40 epochs. These training settings aimed to optimize model performance while preventing overfitting. The model was evaluated after each epoch using the validation set to monitor its performance and to prevent overfitting. The performance of the model was evaluated using accuracy and categorical cross-entropy loss metrics.

Moving to the **Paddy diseases detection system**, we have started to prepare the dataset, The Paddy Doctor dataset contains 16,225 labeled paddy leaf images across 10 classes (9 different paddy diseases and healthy leaves). Our main objective is to develop a deep learning-based model to classify the given paddy leaf images accurately. Therefore, we are proposing a highly effective deep learning model based on the fine-tuned Residual Neural Network (**ResNet34**) architecture.

- **Data Preprocessing**

We started with a preprocessing pipeline that was designed to prepare image data for training and evaluation. The pipeline encompassed several key steps to ensure optimal data representation and facilitate efficient model learning. The dataset was split into separate training and validation sets, which allocated 90% of the data for training and 10% for validation. Furthermore, random shuffling of the was performed to eliminate any potential biases caused by the original order of the data. This randomization step enhances the generalization capabilities of the model by exposing it to a diverse and unbiased distribution of images during training. The preprocessing pipeline also incorporated a series of transformations to standardize and enhance the images. These transformations, encapsulated within the `transform` variable, included center cropping the images to a size of 446 pixels, resizing them to a uniform size of (256, 256) pixels, converting them to tensors and normalizing their pixel values. The normalization step employed specific mean and standard deviation values to align the image data with a standard distribution, facilitating more effective training and convergence of the model.(see **Fig**. 7)

Fig 7. Sample of the augmented images

- **Building & Training the Models**

To train the model, we used the cross-entropy loss function, which is commonly used for multi-class classification problems. The Adam optimizer was employed to optimize the model parameters, with a learning rate of 1e-4. During training, we ran the model for 10 epochs and monitored the validation loss to prevent overfitting. We applied early stopping with a patience of 2 epochs to stop the training process if the validation loss did not improve. Throughout the training process, we collected the training and validation losses as well as the training and validation accuracies.

For The **Crop Recommendation and Support system "IoT-SACRSS"**, we are proposing a support system to recommend the most suitable crop to grow based on IoT sensors data obtained from the fields and to help the farmland owners to have precision agricultural insights. We have developed various machine learning models and selected the one with the highest accuracy. The system relies on a KNN machine learning model that predicts the most suitable crops to grow based on temperature, soil pH, NPK values, humidity, and rainfall rate. The support system provides data-driven recommendations for achieving optimal nutrient and environmental conditions to improve crop yield. It provides the farmer with soil health insights during the whole cultivation season and determines their land's nutrient needs based on the NPK values, as the system recommends a soil enhancer, such as biochar and compost.

We started with a Decision Tree classifier that was trained and evaluated on the given dataset. The dataset was split into training (70%) and testing (30%) sets, and the classifier was trained on the training data using the "entropy" criterion. The accuracy of the classifier was measured using the testing data, and a classification report was generated to assess its performance in terms of precision, recall, F1-score, and support for each class.

Secondly, (KNN) classifier was employed and assessed on the dataset. The classifier was trained using the training data and evaluated on the testing data. The KNN classifier was configured with a value of 9 for the "n_neighbors" parameter, indicating that it considered the labels of the 9 nearest neighbors when making predictions.

And finally, the Random Forest classifier was utilized and evaluated on a given dataset. The classifier was trained using the training data and subsequently tested on the testing data. The Random Forest classifier was instantiated with 20 estimators and a random state of zero. After training the classifier, it was used to predict the target labels for the testing data. The accuracy of the classifier was calculated, and the obtained accuracy score was recorded.

The classifier recommends the top 3 suitable crops to grow based on data obtained from the fields. The support system provides data-driven recommendations to get the optimal nutrients and environmental conditions to improve crop yield. It determines the land's nutrient needs based on the NPK values, the system recommends a customized mix of soil enhancers, such as biochar and compost. Using the proposed algorithms, the system can recommend the needed nutrient amounts based on the number of acres & the NPK values. Entered readings are compared against the ideal benchmarks and margins of 10% are applied to assure an acceptable buffer allowance. Difference between the two would mean deficiency or extreme soil environment conditions and based on it, the algorithm would recommend the needed nutrient amounts, based on the number of acres (see **Algorithm 1 & 2 & 3**).

Algorithm 1 The Nitrogen Requirements

```
Read soil nitrogen readings (Nitrogen Readings).
Prompt the user to input the number of acres (AC).
Set the ideal nitrogen levels (I_N) to 50 ppm.
Set X as the input nitrogen value.
if X is within the range of 0.90 × I_N and I_N then
  Print "Your nitrogen levels are good, no need for additional nitrogen."
else
  Calculate the shortage (SH_N) as (I_N − X).
  Calculate the shortage as a percentage of the ideal concentration of
  nitrogen (Sh_N_Percent = (X/I_N) × 100).
  Calculate the amount of N required (Req_N) as (Sh_N_Percent/100)×
  AC.
  Prompt the user to select preferred nitrogen source (Urea or Compost).
  if Urea is selected then
    Calculate the final required N amount (Req_N_Final) as Req_N × 2.
    Print "Your land requires the addition of Req_N_Final pounds of
    Urea."
  else
    Set Req_N_Final as Req_N.
    Print "Your land requires Req_N_Final pounds of Compost."
  end if
end if
```

Algorithm 2 The Potassium Requirements

```
Read soil Potassium readings (Potassium Readings).
Prompt the user to input the number of acres (AC).
Set the ideal Potassium levels (I_K) to 40 ppm.
Set X as the input Potassium value.
if X is within the range of 0.90 × I_K and I_K then
  Print "Your Potassium levels are good, no need for additional Potas-
  sium."
else
  Calculate the shortage (SH_K) as (I_K − X).
  Calculate the shortage as a percentage of the ideal concentration of
  Potassium (Sh_K_Percent = (X/I_K) × 100).
  Calculate the amount of K required (Req_K) as (Sh_K_Percent/100)×
  AC.
  Prompt the user to select preferred K source (Potassium Sulphate or
  Other).
  if Potassium Sulphate is selected then
    Calculate the final required K amount (Req_K_Final) as Req_K × 2.
    Print "Your land requires the addition of Req_K_Final pounds of
    Potassium Sulphate."
  else
    Set Req_K_Final as Req_K.
    Print "Your land requires Req_K_Final pounds of Other Potassium
    source."
  end if
end if
```

Algorithm 3 The Phosphorus Requirements

```
Read soil Phosphorus readings (Phosphorus Readings).
Prompt the user to input the number of acres (AC).
Set the ideal Phosphorus levels (I_P) to 50 ppm.
Set X as the input Phosphorus value.
if X is within the range of 0.90 × I_P and I_P then
  Print "Your Phosphorus levels are good, no need for additional Phos-
  phorus."
else
  Calculate the shortage (SH_P) as (I_P − X).
  Calculate the shortage as a percentage of the ideal concentration of
  Phosphorus (Sh_P_Percent = (X/I_P) × 100).
  Calculate the amount of P required (Req_P) as (Sh_P_Percent/100) ×
  AC.
  Prompt the user to select Phosphorus source (Super Phosphate or
  Other).
  if Super Phosphate is selected then
    Calculate the final required Phosphorus amount (Req_P_Final) as
    Req_P × 2.
    Print "Your land requires Req_P_Final pounds of Super Phosphate."
  else
    Set Req_P_Final as Req_P.
    Print "Your land requires Req_P_Final pounds of Other Phosphorus
    source."
  end if
end if
```

Moving to the second part of the project, the empowering app, we are proposing the AI-powered Smart Planting Guide. We have worked on identifying the plants type based on flowers and leaves characteristics and features. For the leaf classification model, we have developed a Linear Discriminant Analysis model to identify different types of plants based on leave shape, texture, and margins.

The leave dataset was represented as binary black images against white backgrounds. Three sets of features are also provided per image: a shape contiguous descriptor, an interior texture histogram, and a fine-scale margin histogram. For each feature, a 64-attribute vector is given per sample (see **Fig**. 8).

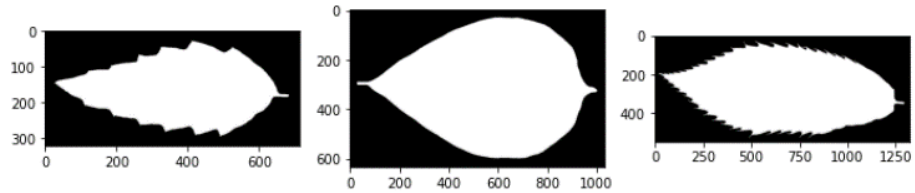


Fig 8. Sample images from the leave images dataset

In agriculture, machine learning classifiers can be used to classify leaves based on their features, such as shape, texture, color, and size. And in our case, which is classifying different types of plants using leaves, we have developed a Linear Discriminant Analysis (LDA) classification algorithm that finds a linear combination of features to separate data into classes. The steps involved are:

- Compute the mean value of each class.
- Compute within-class (S_w) and between-class (S_b) scatter matrices.
- Compute eigenvalues and eigenvectors of Sw^{-1} and Sb to find the directions of projection.
- Project the data onto the new subspace defined by the eigenvectors.

There are two types of scatter matrices that are computed in LDA. The within-class scatter matrix (S_w) measures the variability of the samples within each class, while the between-class scatter matrix (S_b) measures the difference between class means. These matrices are computed as in Eq.(1) & Eq. (2):

$$S_w = \sum_{i=1}^{k} \sum_{x \in C_i} (x - m_i)(x - m_i)^T \quad (1)$$

$$S_b = \sum_{i=1}^{k} n_i (m_i - m)(m_i - m)^T \quad (2)$$

For the flower classification, we have developed a fine-tuned (ResNet50) model and trained it on the flower's dataset . As the dataset was not large enough to train our model, we applied some data augmentation techniques. These techniques include random rotation, horizontal and vertical flipping, affine transformation, color jittering, random resized crop, random crop with padding, Gaussian blur, random perspective transformation, conversion to tensors, and normalization (see **Fig**. 9). These augmentations introduce variations to the images, enhancing the model's ability to generalize and handle diverse input data. By applying these transformations, the robustness and performance of the model trained on the flower dataset has improved.

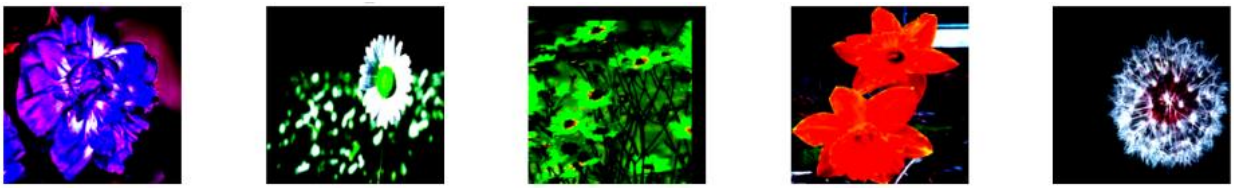


Fig 9. Sample of the augmented flowers images

The flower classification model takes advantage of the pre-trained ResNet50 model and adds a custom fully connected layer to perform classification. The last layer of the model is replaced with a fully connected layer consisting of two linear layers and a dropout layer with a probability of 0.3 to prevent overfitting. The first linear layer takes the output of the model and passes it through 128 nodes. The

second linear layer takes the output of the first linear layer and passes it through the number of classes in the dataset. The model uses the Cross-Entropy loss function and the Adam optimizer. A step scheduler is used for the optimizer.

4.3 Developing the Empowering App & The Smart Farming Website

To utilize these AI models and IoT-SACRSS system, we focused on developing two end-user products, Together, a mobile application to raise awareness among communities about climate change and provide a tool to make planting easier and a web dashboard to provide farms and agricultural domain with comprehensive tools that will enable them to reduce the impact of climate change on crop yields and increase productivity.

We have implemented all deep learning models in Python 3.9 programming language using TensorFlow and PyTorch Libraries on the Kaggle platform using NVIDIA TESLA P100 GPUs. After training the models, we have saved their best weights and developed a lightweight RESTful API using Flask to use our models and make predictions. It was implemented using python 3.8 using the PyCharm IDE and we deployed the backend code using AWS cloud services with EC2 instance. Android Studio has been used to develop the app using Flutter framework and dart language. MongoDB database system has been used to build the plant's information database.

The mobile application offers several features to encourage people to act against climate change. We have developed the app using Flutter as a cross-platform framework, connected to a backend API to receive the images, process them, infer, and return predictions. The app consists of 4 main features which are the AI-powered planting guide that can identify plant species from leaves, and flowers. This will help users to learn more about their plants and make planting easier. We have built our plant information database using MongoDB to provide a step-by-step guide on when to water the plant and all its information about how to care for it . The user can take a picture of his/her plant and the at the cloud will predict the plant type and based on the type; the data will be retrieved from the database. Each user has his own garden where he can add or remove his favorite plants.

The app introduces a community platform where people can share posts, tips, and raise awareness through posts and discussions. Users can upload posts with or without pictures, share tips and facts and interact with each other with likes and comments. Firebase has been used to Authenticate users using google accounts along with Firestore & Fire storage services to handle the user's data and social media platform data. Another feature of the app is the pollution report that will allow users to report polluting activities they observe around them feature will enable people to report polluting actions they see daily around them, such as burning rice straws, by using the user location and phone camera, the user can take a picture of the polluting activity to inform specialized authorities about it. And for the smart farming website, we have implemented the front-end using React Js and JavaScript. It consists of 3 pages, dashboard, weed detection and disease detection.

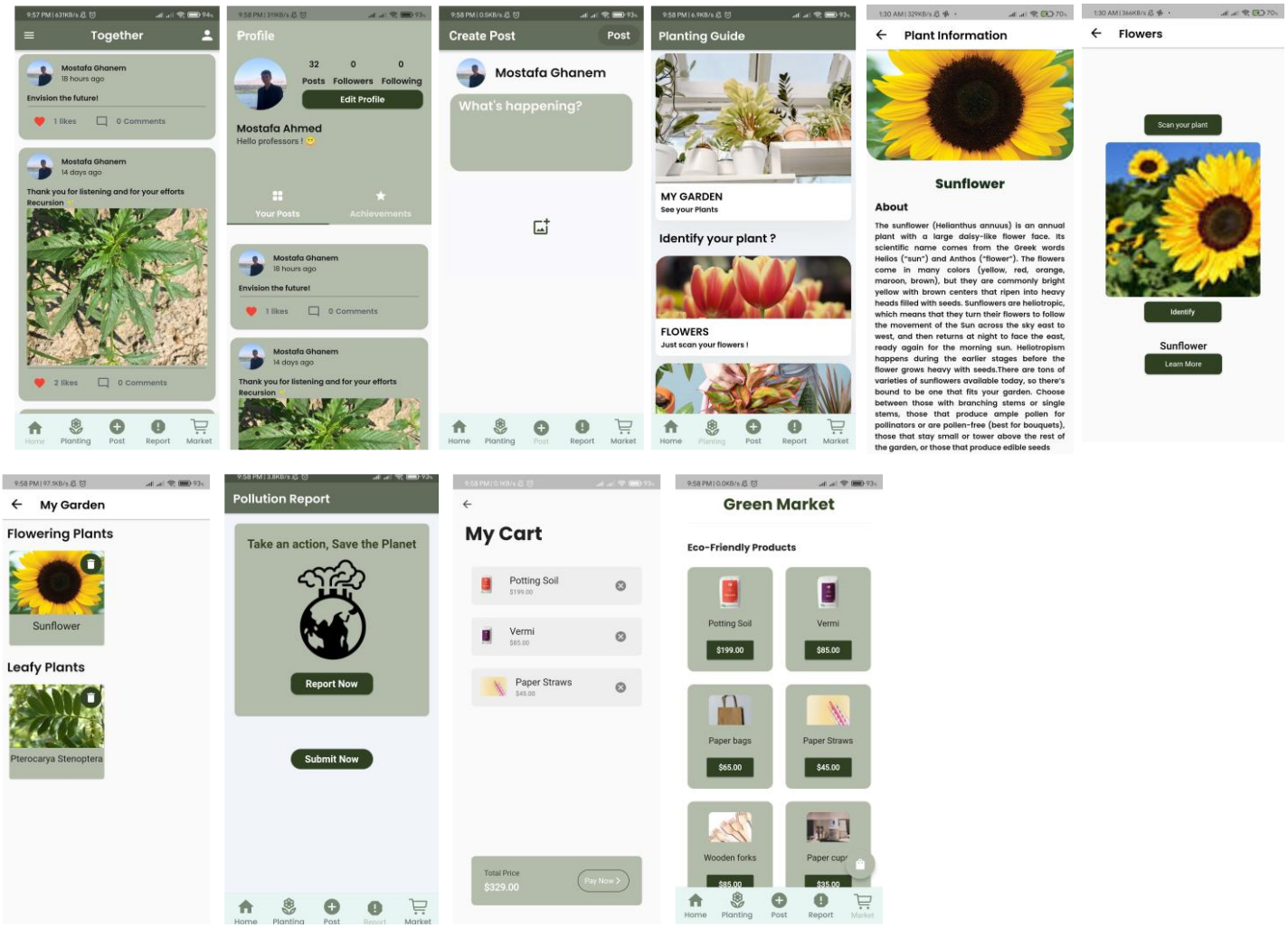
The hardware setup for the Smart Farming System is designed to collect essential agricultural data and transmit it to the central system for analysis and decision-making. The system utilizes an Arduino board, which acts as the main controller, connected to 7sensors for data collection. The specific sensors used in the system are as follows:

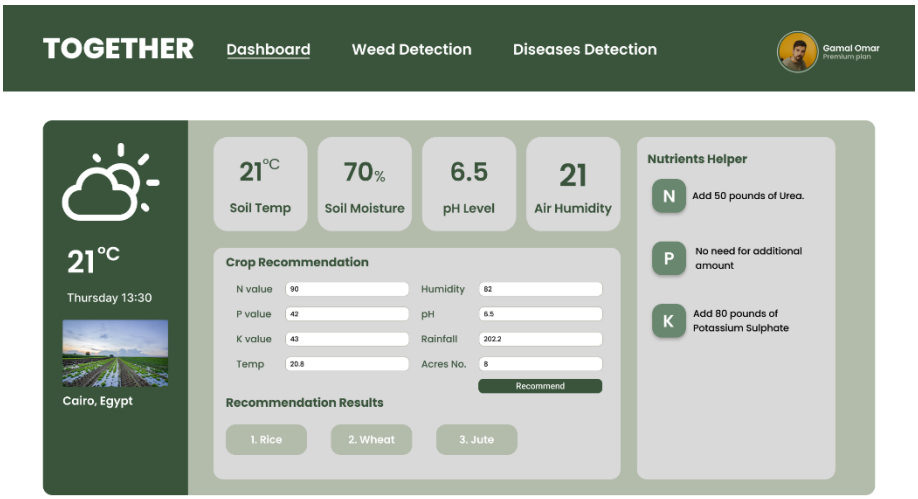
- DHT11 Temperature & Humidity Sensor
- Soil Moisture Sensor
- Soil pH Sensor
- NPK Sensor

The data collected from these sensors is then sent to the Firebase real-time database for storage and retrieval. The Firebase platform provides a scalable and secure cloud-based solution for managing the agricultural data collected by the sensors.

TOGETHER

4.4 User Interfaces





5. Testing and Performance Evaluation

5.1 Weed detection system

ResNet50 Model

Overall, our ResNet50 model achieved the higher accuracy among all the models that we have tried for the weed classification, with an average accuracy of **over 93%** across the 4 folds (see **Table**. 2). Table 3 presents a comparison between the models used in the experiment in terms of accuracy and where the proposed model achieved the highest accuracy across all the models used in the experiment.

Table 3. Validation Accuracy achieved

Model	Valid. Acc. Score
VGG19	0.8467 %
ResNet50	0.9879 %
CNN	0.7946 %

Table 2. Acc. & Loss achieved across folds

Fold	Train Loss	Val Loss	Train Acc.	Val Acc.
1	0.209469	0.191695	0.941731	0.943445
2	0.210408	0.191862	0.936304	0.939160
3	0.216815	0.192072	0.935161	0.943445
4	0.226620	0.215179	0.938018	0.943445

EfficientNetB2 Model

The model building process focused on implementing an efficient model to classify the weeds and crops seedlings with high accuracy. As it was a challenge to extract features from the small seedlings, several deep neural network models, including MobileNetV2, VGG16, and EfficientNetB2 are proposed to classify images into their true classes (see **Table**. 4). The EfficientNetB2 showed the highest accuracy during training and validation with 97 % (see **Fig**. 10). See confusion matrix (see **Fig**. 11).

Table 4. Validation Accuracy scores achieved by each model

Model	Validation Accuracy Score
MobileNetV2	0.9539 %
EfficientNetB2	0.9799 %
VGG16	0.9223 %

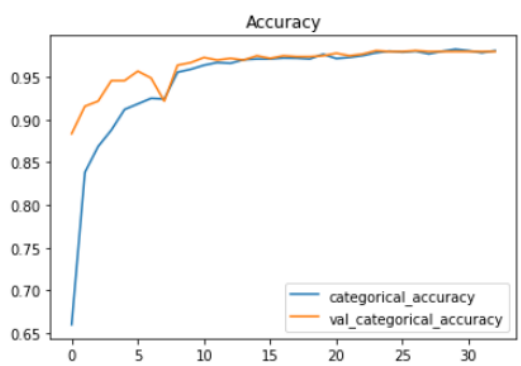


Fig. 10 Accuracy curve for train and validation.

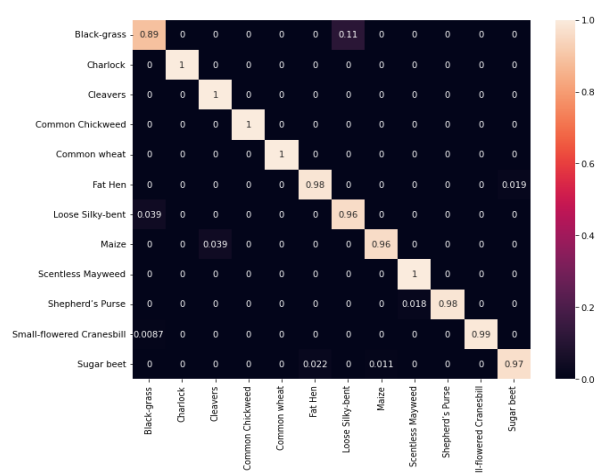
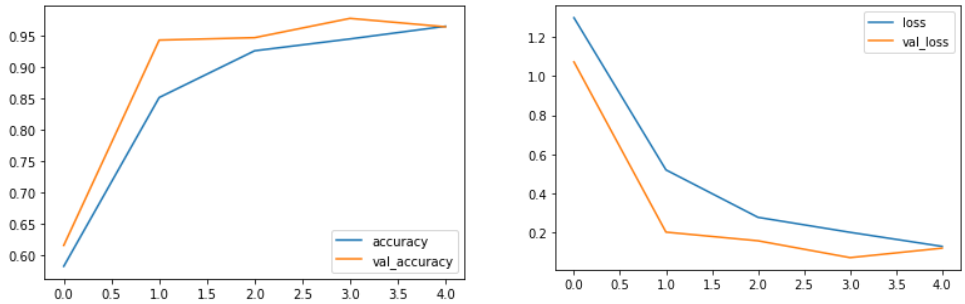


Fig. 11 Confusion matrix of the EfficientNetB2 model

5.2 Paddy diseases detection system (ResNet34)

The model exhibits remarkable capabilities in accurately classifying and identifying 12 distinct paddy leaf diseases. The model showed a high accuracy during training and validation (see **Fig**. 12).



(a) (b)

Fig. 12 Acc. & loss curves. (a) Acc. curve for train & valid. (b) Loss curve for train & validation.

5.3 Crop Recommendation (Random Forest)

The Random Forest Classifier showed the highest accuracy among the other classifiers (see **Fig**. 13).

For evaluation, we have used confusion matrix (see **Fig**. 14).

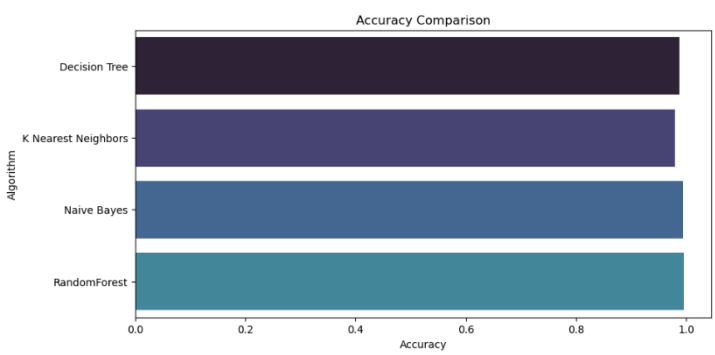


Fig. 13 Acc. comparison between classifiers

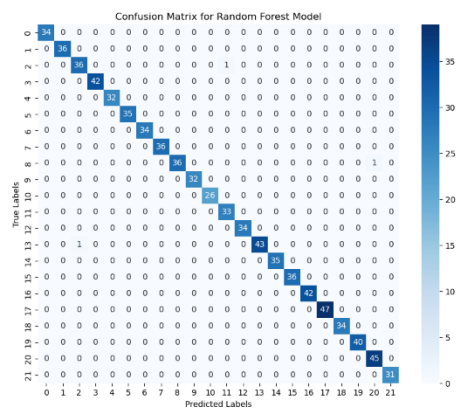


Fig. 14 Confusion matrix of the RF Classifier

5.4 Flower Classification (Resnet50)

Our fine-tuned ResNet50 model achieved the higher accuracy among all the models that we have tried for the weed classification, with an average accuracy of **over 93%** across 4 folds (see Fig. 15).

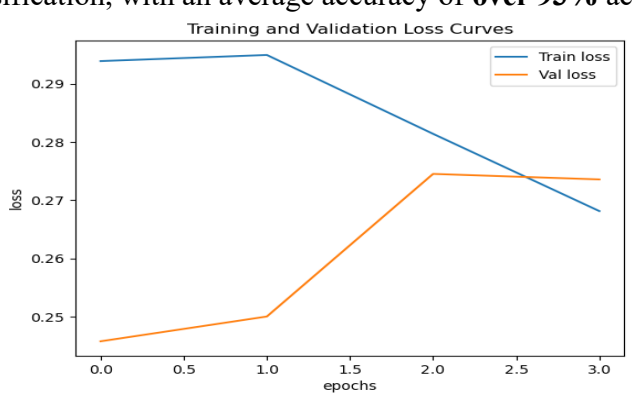


Fig. 15 Accuracy curve for train and validation for the last 3 epochs in the 4th fold

5.5 Leaves Classification (LDA)
The LDA classifier achieved ≅ 0.98 % for validation accuracy (see **Fig**. 16).

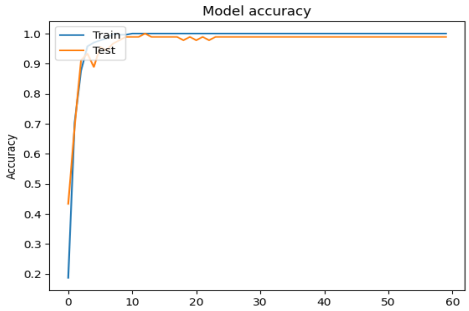


Fig. 16 Accuracy curve for train and validation

6. Usability and Societal Impact

Usability

The Together application is designed with user-friendliness as a top priority, ensuring that individuals from diverse backgrounds can easily navigate and access its features. The user interface is intuitive and straightforward, allowing users to seamlessly engage with the app's various functionalities. The smart planting guide utilizes AI technology to simplify the process of normalizing planting, providing users with personalized recommendations based on their location, climate, and preferences. The community platform fosters a sense of belonging and collaboration, enabling users to share experiences, learn from one another, and collectively act against climate change. Additionally, the pollution reporting feature ensures that reporting environmental incidents is a hassle-free process, encouraging users to play an active role in environmental protection. Through continuous user feedback and updates, the Together app strives to enhance its usability and cater to the evolving needs of its users.

Societal Impact

By raising awareness and understanding of how simple human actions can impact climate change, the app empowers individuals to make informed choices in their daily lives, thus contributing to a collective effort in combating climate change. As users adopt sustainable habits encouraged by the smart planting guide, they directly contribute to reducing carbon footprints and promoting environmental conservation. The community platform fosters a sense of solidarity, inspiring more people to join the movement and work together towards a sustainable future. By encouraging users to celebrate and share their progress and stories, the app amplifies the impact of individual actions, creating a positive ripple effect that can reach a broader audience.

In addition, the application focuses on empowering communities and building resilience against climate change strengthens the collective capacity to adapt and mitigate its impacts on agriculture and food security. The novel smart agriculture system equips farmers with advanced tools to combat the challenges posed by climate change, ultimately enhancing agricultural productivity and sustainability. By providing farmers with intelligent weed and disease detection, along with crop recommendations based on soil analysis and IoT insights, the app supports efficient and environmentally conscious agricultural practices.

7. Conclusions and Recommendations for Future Work

In conclusion, this project proposes an innovative solution to the challenges of climate change and agriculture through the development of the "Together" mobile application. The app aims to create awareness and motivate individuals to take small actions that contribute to reducing climate change. It provides various solutions and habits to monitor while motivating users to continue making positive changes and gaining more awareness through achievable methods and make planting easier.. The proposed smart farming system addresses the challenges of climate change by decreasing GHG emissions and increasing crop productivity through the expansion of woodland and soil and carbon farming. The app employs AI and IoT technologies to provide intelligent weed and disease detection systems and guide crop recommendations based on environmental conditions. The "Together" app offers a unique approach to build a community of people concerned with climate change by focusing on habit-building, empowering communities to increase resilience against climate change. In this way, the app offers a practical and accessible solution that can make a meaningful impact towards achieving sustainable agricultural practices and combating climate change.

Future Work

- Expanding the app's features and functionality: The app can be enhanced by adding new features such as an interactive map to showcase the global impact of climate change, personalized carbon footprint calculators, and gamification elements to the app to make it more engaging.
- Conducting further research and development: Further research can be conducted to improve the accuracy of the AI and IoT technologies used in the smart farming system. This could include developing more advanced sensors and algorithms to improve weed and disease detection and crop recommendations.
- Evaluating the app's impact: It would be valuable to conduct studies to evaluate the effectiveness of the "Together" app in promoting sustainable habits and contributing to climate change mitigation. User feedback can also be collected to improve the app's user experience and identify areas for further improvement.
- Scaling the app and farming system: The app and the smart farming system can be adapted and scaled to cover more plant types and crops.
- Integration of drones and IoT technology for precision farming: In addition to the current farming system features, integrating drone and IoT technology can help farmers to monitor crop health and identify issues such as nutrient deficiencies, pest infestations, and drought stress. The drone can be equipped with sensors and cameras that collect data on soil and crop health, which can then be analyzed using AI algorithms to provide farmers with actionable insights.

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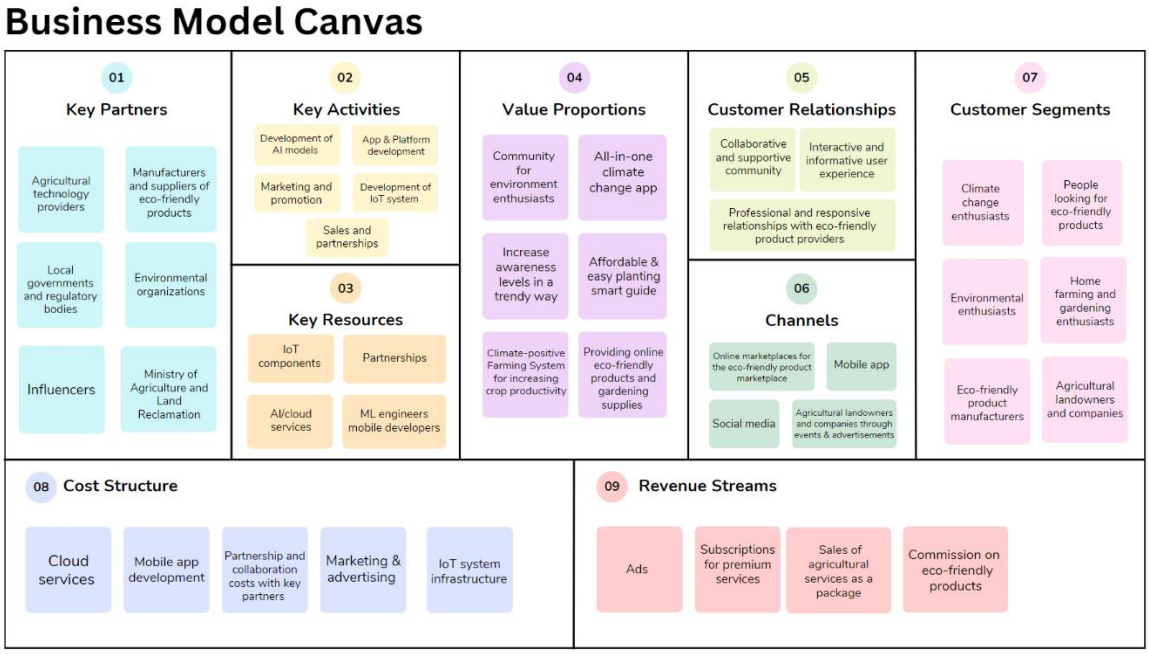
9. Appendix

Appendix A: